

How Predictable Is Pokémon Champions? A Ratings-and-Models Study

Measuring the irreducible variance of a competitive VGC metagame

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*An empirical study, run entirely over ABRA's durable store of real Champions (Reg M-B) ladder games. It asks a question every competitor feels but rarely measures: **when the better team or the better player shows up, how often do they actually win?** The answer bounds what any pre-game model — JOLTEON, a calibrated CHOMP, anything — can achieve, and it reframes the models' "only ~56%" not as weakness but as brushing against a hard ceiling the game itself imposes. Reproduce with `engine/predictability.py`.*

o. The CHOMP–JOLTEON–win% dynamic, in one place

Three things people conflate, separated:

- **JOLTEON predicts the winner.** Given two team sheets, it outputs a calibrated win *probability*. It is a forecaster. Its accuracy is ~55%.
- **CHOMP decides what to do.** Given your six and theirs, it picks the four to bring and two to lead by exact damage. It is a decision engine. It does not forecast a winner at all.
- **Winning percentage is the ceiling both live under.** This study shows the game only lets the better side win ~55%, rising to ~58% with a big edge. That number is set by the game's variance, not by any model.

How they relate:

1. **As predictors, CHOMP and JOLTEON would be near-identical in accuracy (~55–56%)** — because both read only the team sheet, and the sheet's predictive power is capped at the ceiling. They differ in *how* they reach it (JOLTEON = learned species strengths; CHOMP = exact damage), so they make **different mistakes**, which is why you combine them (CHOMP's damage score as a feature in JOLTEON) rather than pick one.
2. **CHOMP raises your win% a different way than JOLTEON measures it.** JOLTEON tells you the pre-game odds; it cannot change them. CHOMP acts *inside* the game — bringing the right four — which moves the part of the outcome the forecaster can't see. So CHOMP improves winning *by deciding well*, not by predicting well.
3. **The ceiling is the whole point.** Because pre-game prediction caps at ~55–58%, the leverage is downstream of the sheet: in-game decisions (CHOMP now; MEDICHAM/SLOWKING deeper) and team building against the meta (DITTO). Prediction is nearly maxed out; *decisions and construction are not*.

The rest of this study is the evidence for that ceiling.

1. Why this matters

Every model in ABRA that predicts a *winner before the game* — JOLTEON today, a calibrated CHOMP tomorrow — lands around 55–57% accuracy, and we have reported that honestly throughout. The natural worry is that the models are simply weak. This study tests the alternative hypothesis: that **55–57% is close to the maximum any pre-game predictor can reach**, because Champions is a high-variance game in

which the better side wins only slightly more than half the time. If true, chasing a team-sheet model to 70% is chasing a number the game does not contain, and the real leverage is elsewhere (in-game decisions, team building against the meta) — exactly where ABRA's other models aim.

We attack it two ways that need no model at all, then check the model against them.

2. Data

All finished games in the durable store (4,998 unique replays). For the ratings analysis we keep the **2,337 human-vs-human games where both players have a Showdown rating**. Ratings are read directly from the replay `|player|` lines; no modelling is involved. For the JOLTEON analysis we use the same humans-only temporal split as the model's own evaluation.

3. Result 1 — how often does the better *player* win?

Across 2,318 games with unequal ratings the higher-rated player won 55.0% of the time — **a figure superseded on 2026-07-25**. It was computed with the NAME-only bot flag, which missed six high-volume accounts. Removing behavioural bots gives **52.4%, 95% CI [49.9, 54.9]**, which includes a coin flip; the full quality filter gives 50.9%. See docs/STUDY-DESIGN-skill-vs-luck.md. Broken down by the size of the rating gap, and compared to what the Elo formula $p = 1/(1+10^{-(\Delta/400)})$ predicts:

Rating gap (Δ)	Games	Higher-rated actually wins	Elo predicts
0–25	592	49.3%	51.8%
25–50	547	54.8%	55.3%
50–100	1034	57.9%	60.5%
100–200	164	57.9%	65.1%

Two things stand out. First, **at small rating gaps the game is a coin flip** — a 0–25 point edge wins 49.3%, statistically indistinguishable from chance. Second, **even a large edge tops out around 58%**, and it does so *below* the Elo prediction: a 100–200 point gap "should" win ~65% but wins only ~58%. Champions compresses skill. The better player is favoured, but the variance — damage rolls, 50/50 protects, speed ties, team-matchup luck — repeatedly overrules them.

4. Result 2 — how well-calibrated is JOLTEON, and where is the ceiling?

On the held-out temporal split (513 games, humans only), JOLTEON scores **54.6% accuracy, Brier 0.252, log-loss 0.699** — right where the ratings result says a pre-game predictor should live. Its probabilities are reasonably calibrated: when it says 55–65%, the favoured team wins ~60%; when it says under 35%, they win ~32%. It is honest about its own uncertainty rather than overconfident.

The telling number is the **ceiling**. JOLTEON's most extreme honest probability on this data is about **81%**, and it reaches that only for a handful of lopsided team-sheet mismatches. For the bulk of real games — two competent teams — it correctly declines to be confident, because the game does not permit confidence. This is the same wall the ratings analysis hit from the other side.

5. Result 3 — CHOMP as a predictor (method and expectation)

CHOMP already computes an exact-damage matchup score (`teamVs` / `bring4`) for the best-4 of one six against the other. Turning that into a win probability is a one-line calibration: $P(\text{win}) = \sigma(a \cdot (\text{score}_1 - \text{score}_2) + b)$, with `a`, `b` fit against the stored outcomes — a logistic on data we hold. `engine/chomp-`

`predict.js` emits the per-game score difference for this fit.

We report the **method** rather than a full backtest number here for an honest reason: computing CHOMP's score requires the full damage engine, and over thousands of games its memory footprint exceeds this analysis sandbox's ceiling (it runs in small batches; a full run needs a roomier machine). But the *expected* result is not in doubt, and Results 1–2 tell us why: a calibrated CHOMP predictor is another **pre-game team-sheet model**, so it is bounded by the same ~55–58% ceiling. Grounding the prediction in exact damage instead of learned species strengths changes *how* it reaches the ceiling, not *where* the ceiling is. The genuinely valuable use of CHOMP's score is therefore not as a rival predictor but as a **feature inside JOLTEON** (§7).

6. Synthesis — the predictability ceiling of Champions

Three independent measurements agree:

- the **better player** wins ~55% (58% with a big rating edge),
- the **better team** (JOLTEON) is called correctly ~55%, calibrated, with a practical confidence ceiling ~81% reached only on rare mismatches,
- a damage-grounded predictor (CHOMP) is bounded by the same wall by construction.

So the answer to "*how often does the better side win?*" is: **only a little more than half — around 55%, rising to ~58–60% when the edge is large**. That is the irreducible variance of the format. It is not a flaw in the models; it is a property of the game, and the models are already near it.

What about the better *play* (per turn)?

This study measures the better *team* and the better *player*. Measuring how often the better *in-game move* wins is harder: it needs a solver that can say which move was better, i.e. SLOWKING's belief search. We cannot yet quantify it from data alone. But the team/player ceiling **bounds** it — in a game where the better player wins 55%, individual correct plays swing games by less, which is exactly why a coach (KADABRA/ALAKAZAM) should talk in terms of *shifting win probability at the margin*, not guaranteeing outcomes.

7. Implications for ABRA

- **Stop chasing pre-game accuracy past ~58%.** It is not there. JOLTEON's rarity-aware ~56% is near-optimal for a team-sheet model; the honest move is calibration, not more accuracy.
- **The leverage is downstream of the sheet.** In-game decisions (MEDICHAM, SLOWKING) and team-building against the meta (DITTO) act on the part of the outcome the sheet cannot see — which is most of it.
- **Feed CHOMP's damage score into JOLTEON as a feature**, not as a competitor. It adds information a learned-identity model lacks, and it is the most principled way to squeeze the last point or two out of the pre-game predictor before the ceiling stops everyone.
- **Coaching should be probabilistic.** Because even large edges only win ~58%, KADABRA's advice is correctly framed as "this play was +X% win probability," not "this play wins."

8. Honest limits

The ratings are Showdown ladder Elo, which is noisy and non-stationary; the humans-only rated sample is 2,337 games from a ~2-day window, so the rating-gap bins (especially 100–200) are small and their confidence intervals are wide. The direction — a low, slowly-rising better-side win rate well under Elo's

prediction — is robust across bins and matches the independent JOLTEON result, but exact percentages will move as the store grows (and the recency-weighted models will track that drift). The CHOMP backtest number is deferred, not claimed.

9. References

1. Elo, A. (1978). *The Rating of Chessplayers, Past and Present*.
 2. ABRA — [simulator white paper](#) (the ~55–57% ceiling, derived); [SLOWKING white paper](#) (why per-move "better play" needs a solver).
 3. Reproduce: `engine/predictability.py` , `engine/chomp-predict.js` .
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